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# Irradiance and solar cycle variability: clues in cycle phase properties

J.R. Kuhn *

*Institute for Astronomy, University of Hawaii, 2680 Woodlawn Avenue, Honolulu, HI 96822, USA*

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**Abstract**

We see solar cycle-related variations in almost every coronal, chromospheric, and photospheric measurement obtained over a several-years timescale. With such an abundant observational foundation, we should expect our physical models of the solar cycle to be overconstrained. In fact this is not the case and the "standard dynamo model" is not predictive. This paper describes some new thoughts on how solar irradiance variations may provide clues to the solar cycle regulation mechanism near the radiative–convection zone boundary.
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*Keywords:* Solar irradiance; Solar cycle variability

## 1. Introduction

An important goal of solar-terrestrial studies is to understand the past and likely future solar inputs to the terrestrial climate. This question is difficult because we lack a fundamental (or predictive) understanding of the solar cycle. It is unclear if secular changes in the Sun are significant or how climate may have been affected. We are stalled by the need to better understand the solar variability mechanisms.

There are many views of the solar cycle and the induced irradiance variation. At one extreme is the belief that the solar cycle is "reasonably well understood" as a dynamo process which manifests several global scale solar effects (cf. Gilman, 2002). An alternative is that current dynamo models are incomplete. How the magnetic solar cycle influences irradiance is also controversial, but the scientific community seems to divide into the "superficialists" who account for and "explain" irradiance variability as being caused by the superficial effects of sunspots, which darken, and faculae, which brighten the photosphere. An alternative is that the irradiance is a more complicated function of the heat transport properties of the convection zone and the magnetic fields both at the tachocline (the boundary region between the solar radiative and convection zones) and the photosphere (Kuhn, 1996).

Much has been written about the complex relationship between surface magnetic fields and irradiance (cf. Kuhn and Armstrong, 2003). Here we reiterate two contrasting mechanisms. In one case photospheric magnetic fields decrease irradiance, while in the other the fields originating from the tachocline must increase luminosity. Thus surface magnetic fields and irradiance changes can be correlated or anticorrelated depending on their latitudinal distribution and whether they originated from the deep interior. The timescale for the irradiance change is also critical to whether the irradiance-magnetic correlation is positive or negative (Kuhn and Armstrong, 2003).

Classical dynamo models may explain the solar cycle magnetic variations. The $\alpha$–$\omega$ and flux transport models (Babcock, 1961), while not yet exhibiting empirical predictive power, seem to be consistent with the global solar cycle magnetic and seismic rotation data (Gilman, 2002). Interestingly, the stochastic flux transport model which Leighton (1969) described decades ago does nearly as well as recent models at reproducing the observable details of the magnetic cycle changes. There are some attractive features in this view which are more persuasive when we consider an observational characteristic of the magnetic solar cycle which is often ignored

* Tel.: +1-808-956-8968; fax: +1-808-956-4604.
*E-mail address:* kuhn@ifa.hawaii.edu (J.R. Kuhn).

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– the phase coherence, but amplitude variability, of the solar magnetic cycle.

## 2. Accurate photometry

### 2.1. Anticorrelated fields and irradiance

Solar photometry is available with sufficient accuracy and spatial resolution that the irradiance contributions from sunspots and faculae can now be directly measured, rather than inferred from "proxy" observations based on, for example, CaIIK measurements (Kuhn et al., 1999). While the approach of estimating solar irradiance from proxies has statistical utility for predicting the Sun's irradiance, the simplifying assumptions which go into two- or three-component models are not physically correct.

For example Fig. 1 shows how faculae are often *dark* at disk center. This plot shows the facular contrast between disk center and the limb as a function of the CaIIK strength of the facular signal. The thicker plotted curves show data for faculae with a stronger CaIIK flux. Note that they also are dark at the disk center in continuum wavelengths (compared to surrounding quiet photosphere). Fainter features (like the network CaIIK Structures) show very little continuum contrast and have a relative brightness which is effectively the same at disk center as near the limb. These measurements were obtained using the Precision Solar Photometric Telescope operating from Mauna Loa. Obviously any statistical model which assumes an irradiance contribution from faculae that can only be bright or which depends on a universal contrast across the disk can have only statistical, but not physical significance.

The net effect of faculae on the irradiance cannot be evaluated from the results of Fig. 1 alone without complete knowledge of the CaIIK intensity distribution and because we know that faculae and spots also affect the surrounding photospheric brightness in a way which is not contained in the local brightness contrast function measurements. Nevertheless these results show that magnetic fields near disk center must decrease the irradiance.

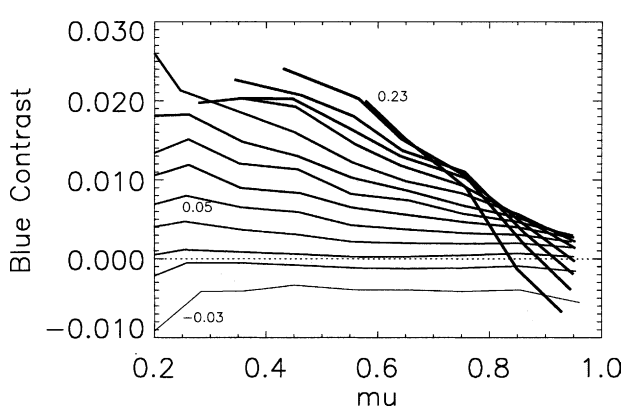


Fig. 1. The facular contrast versus cosine of the central angle is plotted here. Disk center is toward the right. Each curve shows the continuum contrast as a function of the CaIIK contrast of the faculae. Thus the strongest faculae (contrast near 0.23) have the largest contrast, but are *dark* near disk center.

### 2.2. Correlated fields and luminosity

Fields which originate from the solar RZ should have the opposite effect on the solar luminosity. Fig. 2 shows schematically how flux which grows in the RZ, due to the strong rotational shear, is heated more than the ambient non-magnetized fluid. The timescale, $\tau$, for this radiative heating effect depends on $C_p$ the heat capacity, $\kappa$ the opacity, $\rho$ the density, $T$ the temperature, $l$ the magnetized layer vertical thickness, $\mathrm{d}T/\mathrm{d}z$ the vertical temperature gradient, and $\Delta = B^2/8\pi P$ the ratio of magnetic to local gas pressure at the top of the RZ,

$$\tau = \frac{3\pi C_p \kappa \rho l}{16\sigma T^2 \Delta \mathrm{d}T/\mathrm{d}z} = 10^2 l/B^2 \ [\mathrm{yr\,G^2/cm}]. \quad (1)$$

Here $\sigma$ is the Stefan–Boltzmann constant and the final equality gives the heating timescale at the tachocline in terms of the layer thickness (in cm) and field strength (in G).

Despite the relatively large opacity and pressure at the RZ–CZ boundary it is possible to heat the magnetized plasma by an interesting amount. For example, consider the solar cycle luminosity fractional amplitude of $10^{-3}$. Could this luminosity excess be carried by the solar cycle toroidal flux which originates in the tachocline? For this we must have $11\ \mathrm{yr}/\tau \approx 10^{-3}$. We use Parker (1996) estimate of the toroidal flux as $\phi \geqslant 10^{23}\mathrm{Mx} = 5 \times 10^{10} Bl/[\mathrm{Mx/G/cm}]$, and we've assumed a flux sheath that covers 30% of the latitudinal surface area. Combining this estimate of the total flux (which could be much larger in the tachocline) with the required heating rate implies that $B \geqslant 6$ MG, and $l \geqslant 3$ km. Stronger fields and thicker flux regions could also satisfy the physical constraints.

This schematic calculation illustrates an interesting principle – that magnetized plasma which originates from the RZ should be hotter than the unmagnetized surrounding fluid. When such magnetic flux emerges at the photosphere, where the opacity is small, it must increase the solar luminosity. This is independent of how magnetic fields redirect the emergent energy flux at the photosphere (the magnetic anticorrelation effect described above). Thus there should be a tendency for magnetic fields to be hotter and brighter on timescales where this excess entropy generated from the RZ can be radiated out of the magnetized plasma. This generates a positive correlation between surface magnetic fields and the solar luminosity.

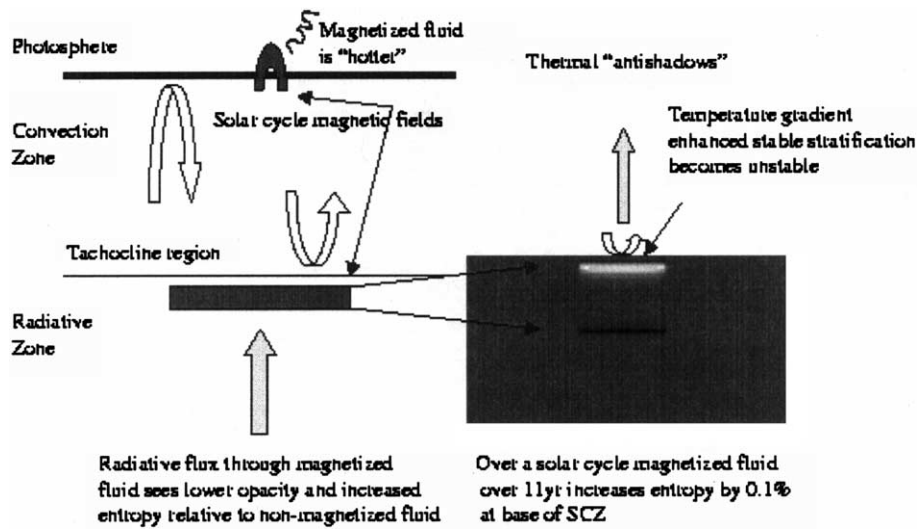


Fig. 2. Magnetic fields trapped in the radiative zone by the rotational shear heat up compared to non-magnetized fluid. A radiative instability drives the initially convectively stable fluid into the convection zone and then to the surface.

Another interesting feature of such an instability at the top of the RZ is that it could serve to regulate the growth and emission of magnetic flux from RZ into the solar CZ. The tachocline is characterized by a nearly adiabatic temperature gradient such that even small perturbations in the local temperature profile, perhaps caused by the heating of magnetized fluid imbedded in the initially stably stratified fluid, could lead to local convective instability. The growth and convective trigger which emits flux from the RZ, while speculative, is a problem amenable (and ripe) for numerical analysis.

## 3. The solar oscillator phase and amplitude properties

The sunspot timeseries record (Fig. 3) has been exhaustively studied. Even a cursory examination suggests an oscillator with a highly variable amplitude and a less changeable period of about 11 years. With a cycle amplitude that varied from zero during the Maunder minimum in the late 17th century and a fluctuating peak amplitude of more than a factor of 2 as seen during the last century, it is obvious that a simple harmonic oscillator falls short of explaining the sunspot cycle.

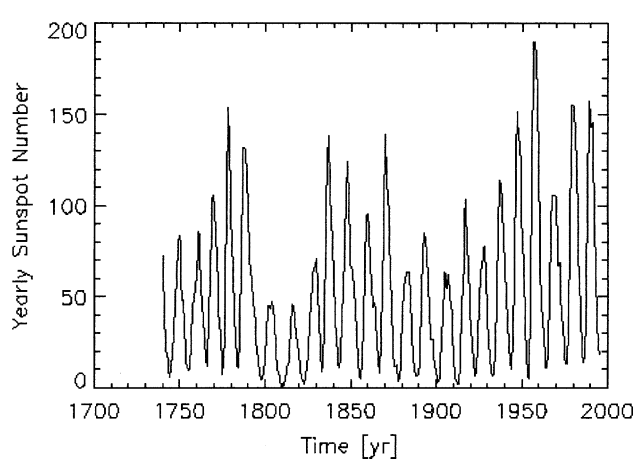


Fig. 3. The average yearly sunspot number record since the Maunder minimum.

In the spirit of the stochastic flux transport model of Babcock and Leighton perhaps we should expect seemingly random amplitude and phase fluctuations. Here the previous cycle toroidal flux randomly arrives at the photospheres from the deep interior and then diffuses from the mid-latitudes to regenerate the global poloidal field. Should not we expect the arrival time and strength of a sunspot peak in one cycle to be only weakly determined by earlier cycles except perhaps for the one immediately preceeding? Certainly there should be no long term phase coherence where an underlying "clock" signal arrives at the surface with only short period fluctuations in the maxima arrival times centered around strictly regular clock beats.

Surprisingly, there is long term coherence to the cycle. The simplest demonstration of this comes from a straightforward Fourier transform of the sunspot record – it reveals a strong and narrow peak near a period of 11 years. This can not occur without phase coherence in the sunspot record. Dicke (1978) may have been the first to do a careful statistical analysis of solar cycle peak statistics, and he showed that there is a high degree of coherence and little evidence of phase wander in the solar cycle record.

The Hilbert transform is an excellent analysis tool for exploring the phase properties of the sunspot timeseries. We compute the Hilbert phase $\phi(t)$ by using Fourier transform techniques to fit the sunspot record $s(t)$ to $A(t)$ exp i $\phi(t)$. We call $\phi(t)$ the "unwrapped" Hilbert phase function. Fig. 4 shows the computed Hilbert phase for the 256 point yearly average sunspot record from Fig. 3.

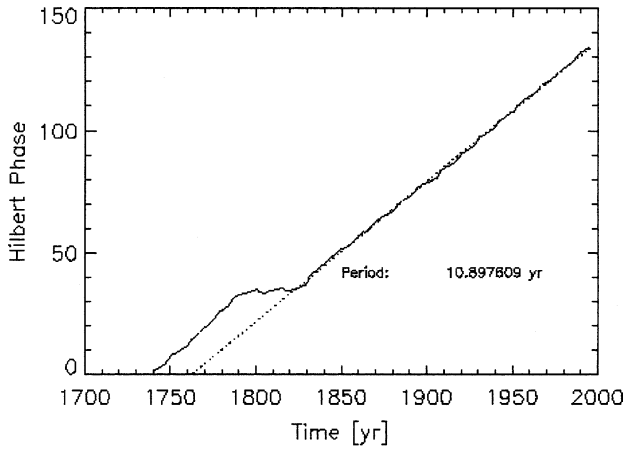


Fig. 4. The Hilbert phase computed from the sunspot timeseries is illustrated here. A linear slope corresponding to a sinusoid of period 10.9 years is indicated with the dotted line.

A surprising result is that, despite the amplitude and fluctuating period between maxima in the timeseries, the Hilbert phase is quite linear with time. It has an accurately defined slope corresponding to a sinusoidal period of 10.898 years. The phase linearity (or period) before 1780 and after 1830 are the same. In fact, the solar cycle phase is remarkably linear except for an interruption near 1800. This is a striking clue to the solar cycle mechanisms which is a distinguishing characteristic which any viable physical model must account for.

### 3.1. Non-linear chaos or stochastically driven?

Many quantitative models of the solar cycle have been inspired by the non-linear dynamo equations (cf. Priest, 1984). The analytic behavior of the sunspot timeseries is indeed consistent with some properties of a non-linear oscillator. Hoyng (1996) successfully related the statistical amplitude-phase properties of the cycle to simplified mean-field dynamo behaviour. A recent relaxation oscillator model using the van der Pol equation (Mininni et al., 2002) reproduces the tendency for a rapid rise and slow decline within each cycle. In contrast Leighton (1969) reproduced many of the statistical cycle properties (like the anticorrelation between cycle duration and amplitude) by stochastically varying the parameters in what could, in principle, even be a linear differential equation for the solar cycle. We consider here a simple schematic model which reflects some of the physical mechanisms and which illustrates how a class of linear models can reproduce the phase properties of the cycle.

### 3.2. The delayed oscillator model

The apparent intermittency of the sunspot cycle during times when few, if any, spots were recorded (like the Maunder minimum period) is sometimes taken as evidence of a strongly non-linear driving oscillation mechanism. We illustrate here how this behavior is also a feature of what we call a "delayed oscillator." It can be illustrated conceptually by considering a simplified picture of flux recirculation between the solar convection (CZ) and the interior radiative zones (RZ). We suppose that at some time $t$ the total flux in the CZ is $F(t)$ while $G(t)$ described the RZ magnetic flux. Motivated by the mechanism described above we assume that flux accumulates and is stored in the RZ and is then emitted into the CZ in proportion to the amount of stored flux at an earlier time. Thus $\mathrm{d}G_{\mathrm{emit}}(t) = \alpha G(t-\epsilon)\mathrm{d}t$ where $\epsilon$ is the storage time and $\alpha$ described the strength of the flux expulsion process.

Similarly we assume that the CZ flux (primarily toroidal) diffuses into a net poloidal field which is differentially wound into a toroidal RZ flux by the Babcock–Leighton diffusion-differential rotation (flux-transport) dynamo. The accumulation of CZ flux absorbed by the RZ is then $\mathrm{d}G_{\mathrm{abs}}(t) = \beta F(t-\delta)\mathrm{d}t$ where $\delta$ describes the storage time in the CZ before toroidal flux regenerates poloidal field lines and $\beta$ accounts for the strength of this process. Since $\mathrm{d}G/\mathrm{d}t = \mathrm{d}G_{\mathrm{abs}}/\mathrm{d}t - \mathrm{d}G_{\mathrm{emit}}/\mathrm{d}t$ and $\mathrm{d}G/\mathrm{d}t = -\mathrm{d}F/\mathrm{d}t$ we can write a single first order differential equation for the solution pair

$$\mathrm{d}[F,G]/\mathrm{d}t = [\alpha G(t-\epsilon) - \beta F(t-\delta), \beta F(t-\delta) - \alpha G(t-\epsilon)]. \tag{2}$$

For small RZ storage and CZ diffusion/winding delays this equation can have oscillatory solutions for $F$ and $G$ with angular frequency $\omega$ given by

$$\omega^2 = \frac{\alpha+\beta}{\alpha\epsilon^2+\beta\delta^2}. \tag{3}$$

Solutions to the delayed oscillator equation for interesting parameter choices tend to yield a growing or damped sinusoid but to see exact solutions it is straightforward to integrate this equation, picking initial conditions for $[F,G]$ and for the four parameters.

More interesting is the behavior of the solutions when the flux emission and absorption rates vary stochastically. The fluctuating convection zone determines $\alpha$ and $\beta$. A fourth-order Runge Kutta solution is illustrated in Fig. 5. The amplitude scale is arbitrary but could be interpreted as the signed toroidal flux. The time scale is also arbitrary and a function of the parameters. Fig. 6 plots the stochastic variability applied to the flux storage parameter $\delta$ and the amplitude parameters $\alpha$ and $\beta$. The stochastic variability has been generated with a temporal correlation of 10 time units – much smaller than the 50 unit average period of the oscillation.

The Hilbert phase is indicated by the dotted line in Fig. 5 and it shows the same behavior we observed in the sunspot record. The phase interruption between 1500 and 2000 time units qualitatively reproduces the phase

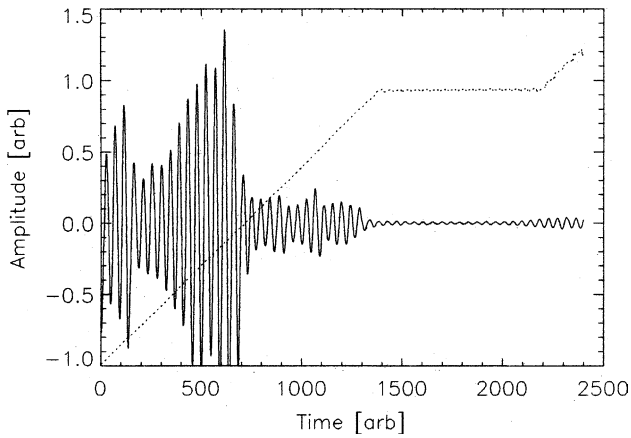


Fig. 5. A sample solution to the delayed oscillator. The toroidal field proxy $F(t)$ is shown by the solid line and the dotted line shows the Hilbert phase (in arbitrary units).

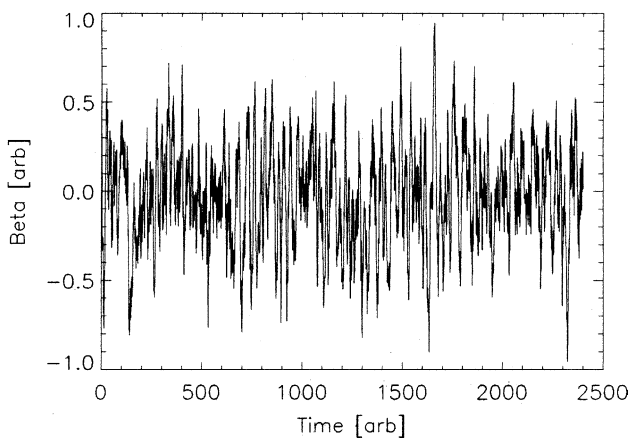


Fig. 6. The fractional "noise" imposed on the delayed oscillator $\beta$ parameter is plotted here.

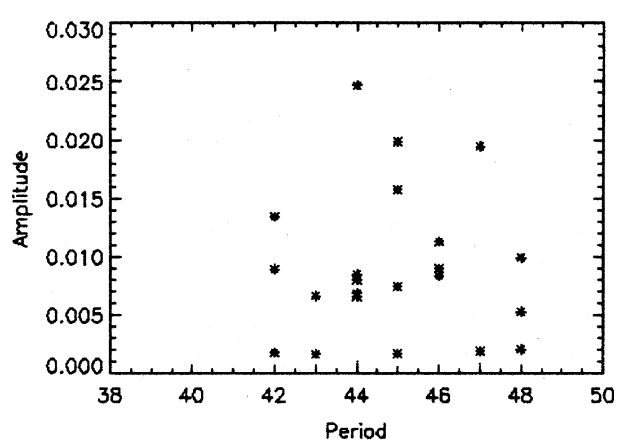


Fig. 7. The period–amplitude correlation from the above delayed oscillator realization. Depending on initial conditions this can be either a positive or negative correlation trend.

interruption in the early nineteenth century sunspot record. As in the sunspot record, the frequency defined by the linear temporal change in the Hilbert phase is constant outside of the phase interruption. Note that the phase stability is not disrupted by the stochastic variability introduced into the parameters, although the oscillation amplitude varies greatly. Fig. 7 shows how the period and amplitude of individual cycles are correlated during the timeseries. This shows how the period changes little as the amplitude of successive cycles fluctuates. We find that the period–amplitude correlation can be positively or negatively correlated over large simulated timescales.

## 4. Conclusions

Sunspots and active region faculae do not explain the observed irradiance variations over the solar cycle. We have argued that one of its most interesting properties may be its coherence, for example, as determined from its Hilbert phase. We also show how a stochastically driven "relaxation, delayed oscillator" has qualitatively similar coherence properties. One way to generate such an oscillation is through a radiative instability at the tachocline whose effect may also be seen in solar luminosity changes correlated with the photospheric magnetic flux.

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## References

Babcock, H.W. The topology of the Sun's magnetic field and the 22-year cycle. Astrophys. J. 133, 572–587, 1961.

Dicke, R.H. Is there a chronometer hidden deep in the sun. Nature 276, 676–680, 1978.

Gilman, P. Observational constraints on the solar dynamo, in: Wilson, A. (Ed.), From Solar Min to Max: Half a Solar Cycle with SOHO. ESA, Noordwyk, The Netherlands, pp. 25–31, 2002.

Hoyng, P. Is the solar cycle tuned by a clock? Sol. Phys. 169, 253–264, 1996.

Kuhn, J.R. Global changes in the Sun, in: Roca Cortes, T., Sanchez, F. (Eds.), The Structure of the Sun: VI Winter School at Instituto d'Astrophysica de Canarias. Cambridge University Press, Cambridge, UK, pp. 231–278, 1996.

Kuhn, J.R., Armstrong, J.D., Mechanisms of solar irradiance variations, in: Pap, J.M. et al. (Ed.), Solar Variability and its Effect on the Earth's Atmosphere and Climate System, American Geophysical Union, 2003 (in press).

Kuhn, J.R., Lin, H., Coulter, R. What can irradiance measurements tell us about the solar magnetic cycle. Adv. Space Res. 24 (2), 185–194, 1999.

Leighton, R. A magneto-kinematic model of the solar cycle. Astrophys. J. 156, L1–L26, 1969.

Mininni, P.D., Gomez, D., Mindlin, G.B. Instantaneos phase and amplitude correlation in the solar cycle. Sol. Phys. 208, 167–179, 2002.

Parker, E.N. The solar magnetic field, in: Roca Cortes, T., Sanchez, F. (Eds.), The Structure of the Sun: VI Winter School at Instituto d'Astrophysica de Canarias. Cambridge University Press, Cambridge UK, pp. 299–352, 1996.

Priest, E.R. Solar Magnetohydrodynamics. Riedel, Holland, 1984.